SC7 Bayes Methods MT26

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Lecture 12: Reversible Jump MCMC I.*

*See Givens and Hoeting, *Computational Statistics*, Wiley, (2013) for a clear introduction and Robert and Casella, *Monte Carlo Statistical Methods*, Springer (2004) for something more like the following.

What problem does RJMCMC solve?

In the joint distribution of the model and the parameter

$$\pi(\theta, m|y) \propto p(y|\theta, m)\pi(\theta|m)\pi(m), \quad \theta \in \Omega_m,\ m \in \{1, 2, ..., M\}$$

the dimension of the parameter θ may vary depending on the model. For eg

$$\begin{array}{llll} m = 1 & Y = \alpha + \epsilon & \theta = \alpha & \Omega_1 = \mathbb{R} \\ m = 2 & Y = \alpha + \beta x + \epsilon & \theta = (\alpha, \beta) & \Omega_2 = \mathbb{R}^2 \end{array}$$

and

$$(\theta, m) \in \Omega^*, \quad \text{where} \quad \Omega^* = (\Omega_1 \times \{1\}) \cup (\Omega_2 \times \{2\}).$$

"The number of things we dont know is one of things we dont know". The MCMC algorithm must jump from between spaces of different dimension.

Reversible Jump MCMC - a first look

Specify $(\theta, m) \to (\theta', m')$; let $Q(\theta', m'|\theta, m)$ be the product of probabilities for the random steps to (θ', m') from (θ', m') and*

$$\alpha(\theta', m'|\theta, m) = \min\left\{1, \frac{\pi(\theta', m'|y)\, Q(\theta, m|\theta', m')}{\pi(\theta, m|y)\, Q(\theta', m'|\theta, m)}\right\}$$

Example: model averaging in regression (again): Model $m \in \mathcal{M}$ is any subset of $[p] = \{1, ..., p\}$; parameter $\theta \in \mathbb{R}^k$ when $|m| = k$.

$$\pi(\theta, \sigma, m|y) \propto p(y|\theta, \sigma, m)\pi(\theta)\pi_s(\sigma)\pi_M(m),$$

with $\pi(\theta|m) = \prod_{i\in m} \pi(\theta_i)$ and $\pi_M(m) = \xi^k(1-\xi)^{p-k}$ and

$$p(y|\theta, \sigma, m) = N(y; X_{:,m}\theta, \sigma^2 I_n).$$

If the model is $m = (i_1, \ldots, i_k)$, we regress on covariates $X_{:,m} = (X_{i_1}, \ldots, X_{i_k})$ with effects $\theta = (\theta_{i_1}, \ldots, \theta_{i_k})$.

*This is actually a special case of RJ which often holds. We give the general formula below. See notes Section 9.5/Appendix A for conditions and proof.

Update mechanism

1. Propose $m \to m'$ by tossing a coin; if heads then propose to add a component and otherwise delete one (if $m = \emptyset$ cant delete and if $m = [p]$ cant add, so reject).

2(up) If adding choose component $i \sim U([p] \setminus m)$ to add;

- Propose new parameter value $\theta_i' \sim \pi(\theta_i')$ (the prior say);
- Set $m' = m \cup \{i\}$ and $\theta' = \theta \cup \{\theta_i'\}$

Note: m' and θ' are ordered - put i and θ_i' in correct positions.

2(down) If deleting select component $i \sim U(m)$ to delete;

- Set $m' = m \setminus \{i\}$ and $\theta' = \theta \setminus \{\theta_i\}$.

3. Probs for event sequences $(\theta, m) \to (\theta', m')$ when $|m| = k$:

$$Q(\theta', m'|\theta, m) = \begin{cases} 1/2 \times 1/(p-k) \times \pi(\theta'_i) & \text{if add,} \\ 1/2 \times 1/k & \text{if delete,} \end{cases}$$

Going back $|m'| = k+1$ if added, $|m'| = k-1$ if deleted:

$$Q(\theta, m|\theta', m') = \begin{cases} 1/2 \times 1/(k+1) & \text{reverse is delete,} \\ 1/2 \times 1/(p-k+1) \times \pi(\theta_i), & \text{reverse is add.} \end{cases}$$

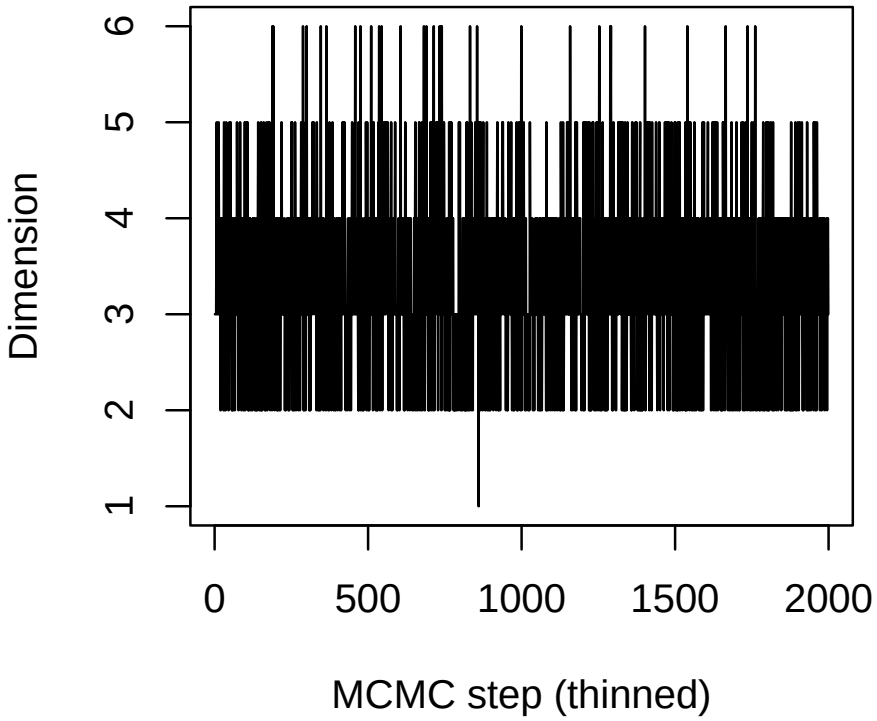
Acceptance probabilities: ($|m'| = k+1$, add)

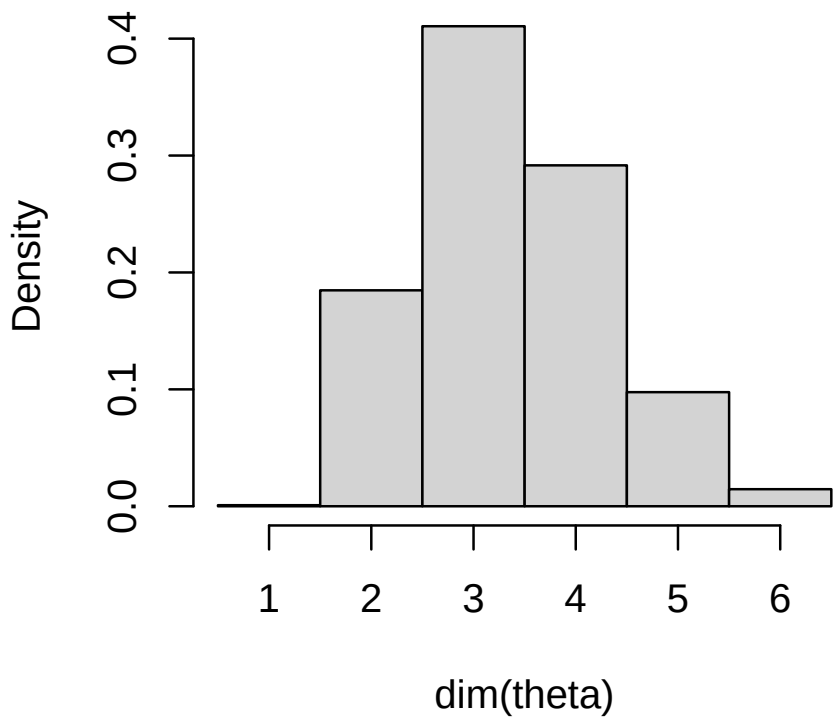
$$\alpha(\theta', m'|\theta, m) = 1 \wedge \frac{\pi(\theta', \sigma, m'|y) \times 1/2 \times 1/(k+1)}{\pi(\theta, \sigma, m|y) \times 1/2 \times 1/(p-k) \times \pi(\theta'_i)},$$

Acceptance probabilities: ($|m'| = k-1$, delete)

$$\alpha(\theta', m'|\theta, m) = 1 \wedge \frac{\pi(\theta', \sigma, m'|y) \times 1/2 \times 1/(p-k+1) \times \pi(\theta_i)}{\pi(\theta, \sigma, m|y) \times 1/2 \times 1/k}.$$

Linear regression of **swiss**-data, random dimension.





See code for this lecture. Model is same as spike and slab model.

For eg. posterior for $|m|$ (here) and $\sum_k z_k$ (there) are equal.

Reversible Jump - a deeper dive

Say $X_t = (\theta, m)$ in MCMC targeting model-averaging posterior

$$\pi(\theta, m|y) \propto p(y|\theta, m)\pi(\theta|m)\pi(m),\ \theta \in \Omega_m,\ m \in \mathcal{M}$$

with $\dim(\Omega_m)$ varying with m. Proposal matching handles this.

(1) Propose $m' \sim (\rho_{m,m'})_{m' \in \mathcal{M}}$.[†]. We now know θ, m, m'.

(2) Build the candidate θ':

- Simulate $u \sim g_{m,m'}(\cdot),\ u \in \mathcal{U}_{m,m'}$
- Set $\theta' = \psi_1(\theta, u)$ (should be $\psi_1^{m,m'}$ etc)
- require $\theta = \psi_1(\psi_1(\theta, u), u')$ for $u' \in \mathcal{U}_{m',m}$ so $u' = \psi_2(\theta, u)$.

[†] $\sum_{m' \in \mathcal{M}} \rho_{m.m'} = 1$ and $\rho_{m,m'} > 0 \Leftrightarrow \rho_{m',m} > 0$

Involution encodes matched proposals for $m \to m'$ and $m' \to m$.

$$(\theta', u') = \psi(\theta, u)$$

with $\psi(\theta, u) = (\psi_1(\theta, u), \psi_2(\theta, u))$ a differentiable involution.

ψ maps $\Omega_m \times \mathcal{U}_{m,m'}$ invertibly onto $\Omega_{m'} \times \mathcal{U}_{m',m}$ and

$$(\theta, u) = (\psi_1(\theta', u'), \psi_2(\theta', u'))$$

so ψ maps

$$\Big(\Omega_m \times \mathcal{U}_{m,m'}\Big) \;\cup\; \Big(\Omega_{m'} \times \mathcal{U}_{m',m}\Big)$$

invertibly onto itself.

The RJ-MCMC algorithm

For each $m, m' \in \mathcal{M}$ such that $\rho_{m,m'} > 0$ let

$$\psi : \Omega_m \times \mathcal{U}_{m,m'} \to \Omega_{m'} \times \mathcal{U}_{m',m}$$

be an involution which is invertible and differentiable.

Suppose $X_t = (\theta, m)$. The state X_{t+1} is determined as follows.

[1] Sample $m' \sim \rho_{m,m'},\ m' \in \mathcal{M}$. Simulate $u \sim g_{m,m'}(\cdot)$.

[2] Set $(\theta', u') = \psi(\theta, u)$ and

$$\alpha(\theta', m'|\theta, m) = \min\left\{1, \frac{\pi(\theta', m'|y)\rho_{m',m}g_{m',m}(u')}{\pi(\theta, m|y)\rho_{m,m'}g_{m,m'}(u)}J_\psi(\theta, u)\right\}$$

[3] With probability $\alpha(\theta', m'|\theta, m)$ set $X_{t+1} = (\theta', m')$ and otherwise set $X_{t+1} = (\theta, m)$.

The Jacobian is non-singular and the update satisfies detailed balance (omitting $X_t = X_{t+1}$)

$$\int_B \pi(d\theta', m'|y)\rho_{m',m} \int_A q_{m',m}(d\theta|\theta')\alpha(\theta, m|\theta', m') =$$
$$\int_A \pi(d\theta, m|y)\rho_{m,m'} \int_B q_{m,m'}(d\theta'|\theta)\alpha(\theta', m'|\theta, m)$$

for all sets A and B for which integrals are defined.

Proof: The proposals are

$$q_{m,m'}(d\theta'|\theta) = g_{m,m'}(u)du$$

and for the reverse update it is

$$q_{m',m}(d\theta|\theta') = g_{m',m}(u')du'.$$

Suppose $\alpha(\theta', m'|\theta, m) = \min\{1, r\}$ with $r \leq 1$ (WLOG).

Under the integrals on the RHS of DB

$$\text{RHS} = \pi(\theta, m|y)\rho_{m,m'}g_{m,m'}(u)\frac{\pi(\theta', m'|y)\rho_{m',m}g_{m',m}(u')}{\pi(\theta, m|y)\rho_{m,m'}g_{m,m'}(u)}\left|\frac{\partial(\theta', u')}{\partial(\theta, u)}\right| d\theta du$$

$$= \pi(\theta', m'|y)\rho_{m',m}g_{m',m}(u')d\theta' du'$$

$$= \pi(d\theta', m'|y)\rho_{m',m}q_{m',m}(d\theta|\theta')$$

$$= LHS$$

using $\alpha(\theta, m|\theta', m') = 1$ for the last step (which holds by same reasoning as before: $J_\psi(\theta, u) = J_\psi^{-1}(\theta', u')$ for an involution).

The transformations are all invertible so when we transform $d\theta d\theta' \to d\theta du \to d\theta' du' \to d\theta' d\theta$ we end up integrating over the same sets we started with in the opposite order. $\square$

Example: general case applied to **swiss** data analysis

$X_t = (\theta, m)$ with $m = (i_1, \ldots i_k)$ and $\theta = (\theta_{i_1}, \ldots, \theta_{i_k})$.

Add a component with probability $1/2$ and otherwise delete one.

[add] $\rho_{m,m'} = 1/2 \times 1/(p-k)$, $u \sim g_{m,m'}(\cdot)$, $g_{m,m'}(u) = \pi(u)$;
Set $\theta'_i = u$ and $\theta'_{-i} = \theta$.

Here $\Omega_m = \mathbb{R}^k$, $\mathcal{U}_{m,m'} = \mathbb{R}$, $\Omega_{m'} = \mathbb{R}^{k+1}$ and

$$\psi_1^{m,m'}(\theta, u) = (\theta_{i_1}, \ldots, \theta_{i_j}, u, \theta_{i_{j+1}}, \ldots, \theta_{i_k})$$
$$\psi_2^{m,m'}(\theta, u) = \emptyset. \qquad (u' = \emptyset)$$

[del] $\rho_{m,m'} = 1/2 \times 1/k$, $u \sim g_{m,m'}(\cdot)$, $g_{m,m'}(u) = \mathbb{I}_{u=\emptyset}$ and

$$\psi_1^{m,m'}(\theta, \emptyset) = (\theta_{i_1}, \ldots, \theta_{i_k}) \setminus \{\theta_i\} \qquad (\text{remove } \theta_i)$$
$$\psi_2^{m,m'}(\theta, \emptyset) = \theta_i.$$

Here $\mathcal{U}_{m,m'} = \{\emptyset\}$.

Acceptance probability [add] using RJ-algorithm: reverse move $m' \to m$ from $k+1$ to k components so $\rho_{m',m} = 1/2 \times 1/(k+1)$.

$$\alpha(\theta', m'|\theta, m) = \min\left\{1, \frac{\pi(\theta', \sigma, m'|y)\rho_{m',m}g_{m',m}(u')}{\pi(\theta, \sigma, m|y)\rho_{m,m'}g_{m,m'}(u)} J_\psi(\theta, u)\right\}$$

$$= \min\left\{1, \frac{\pi(\theta', \sigma, m'|y) \times 1/2(k+1) \times 1}{\pi(\theta, \sigma, m|y) \times 1/2(p-k) \times \pi(\theta_i')} \times 1\right\}$$

where $J_\psi(\theta, u) = 1$ as θ' and θ, u are mapped 1-1 so every row and column has exactly one 1 and the rest 0. Matches previous.

Notice $\dim(\Omega_m \times \mathcal{U}_{m,m'}) = k + 1 = \dim(\Omega_{m'} \times \mathcal{U}_{m',m})$.

I leave it to you to check deletion.

For an example where both $g_{m,m'}$ and $g_{m',m}$ are continuous see last question of PS4.